

Machine Learning Task 1:

Python Code:

Step 1: Libraries import karo

import pandas as pd

import numpy as np

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

Step 2: load Dataset

data = pd.read_csv("/content/sample_data/train.csv")

select Required columns

data = data[['GrLivArea', 'BedroomAbvGr', 'FullBath', 'SalePrice']]

Step 3: handling Missing values

data = data.dropna()

Step 4: separating Features and targets

X = data[['GrLivArea', 'BedroomAbvGr', 'FullBath']]

y = data['SalePrice']

Step 5: Train-Test split

X_train, X_test, y_train, y_test = train_test_split(
 X, y, test_size=0.2, random_state=42
)

Step 6: Linear Regression model

model = LinearRegression()

model.fit(X_train, y_train)

Step 7: Predictions

y_pred = model.predict(X_test)

Step 8: Model Evaluation

mae = mean_absolute_error(y_test, y_pred)

mse = mean_squared_error(y_test, y_pred)

r2 = r2_score(y_test, y_pred)

print("Mean Absolute Error:", mae)

print("Mean Squared Error:", mse)

print("R2 Score:", r2)